Lab 3 - Routing Protocol

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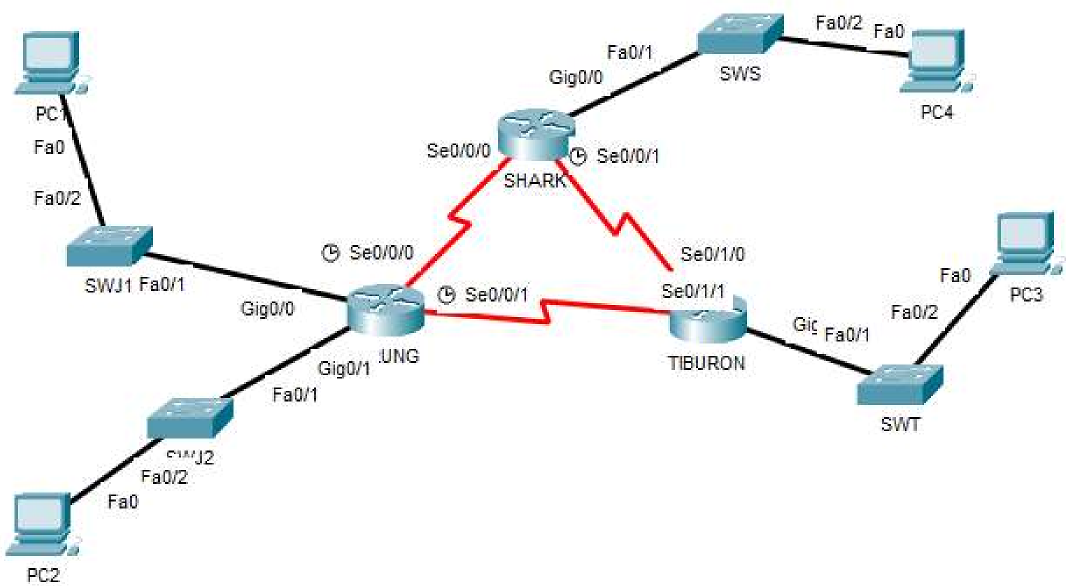
SECTION: 02

INTRODUCTION

This lab looks into the network layer, focusing on subnetting and routing.

TOPOLOGY



LAB INFORMATION

Here is some basic information for the lab. The network address given is 172.18.110.0/23.

Table 1

Device	Subnetwork	Usable Hosts
JERUNG	LAN1	20
	LAN2	30
SHARK	LAN	60
TIBURON	LAN	50
Connections	JERUNG-SHARK	2
	JERUNG-TIBURON	2
	SHARK-TIBURON	2

Table 2

#	Device Name	Interface	IP Address	Subnet Mask	Gateway
1	JERUNG	Se0/0/0	172.18.110.193	255.255.255.252	
2		Se0/0/1	172.18.110.197	255.255.255.252	
3		G0/0	172.18.110.1	255.255.255.224	
4		G0/1	172.18.110.33	255.255.255.224	
5	TIBURON	Se0/1/1	172.18.110.198	255.255.255.252	
6		Se0/1/0	172.18.110.202	255.255.255.252	
7		G0/0	172.18.110.129	255.255.255.192	
8	SHARK	Se0/0/0	172.18.110.194	255.255.255.252	
9		Se0/0/1	172.18.110.201	255.255.255.252	
10		G0/0	172.18.110.65	255.255.255.192	
11	PC1	-	172.18.110.30	255.255.255.224	172.18.110.1
12	PC2	-	172.18.110.62	255.255.255.224	172.18.110.33
13	PC3	-	172.18.110.190	255.255.255.192	172.18.110.129
14	PC4	-	172.18.110.126	255.255.255.192	172.18.110.65

LAB TASKS

Task 1 – IP Addressing

1. Given the network address of the organisation and the basic information provided in both Tables 1 and 2. Show your workings here and complete Table 2 with the correct information. PCs will be given the last usable address of the subnetwork.

JERUNG LAN1	
Required Ips	20 IPs
Subnet Size (2^n)	$2^5 = 32 > 20+2$
Subnet Mask	/32 – 5 = /27
Network Address	172.18.110.0
Broadcast Address	172.18.110.31
Usable IP Range	172.18.110.1 – 172.18.110.30

JERUNG LAN2	
Required Ips	30 IPs
Subnet Size (2^n)	$2^5 = 32 > 20+2$
Subnet Mask	/32 – 5 = /27
Network Address	172.18.110.32/27
Broadcast Address	172.18.110.63
Usable IP Range	172.18.110.31 – 172.18.110.62

SHARK LAN	
Required Ips	60 IPs
Subnet Size (2^n)	$2^6 = 64 > 60+2$
Subnet Mask	/32 – 6 = /26
Network Address	172.18.110.64/26
Broadcast Address	172.18.110.127
Usable IP Range	172.18.110.65 – 172.18.110.126

TIBURON LAN	
Required Ips	50 IPs
Subnet Size (2^n)	$2^6 = 64 > 60+2$
Subnet Mask	/32 – 6 = /26
Network Address	172.18.110.128/26
Broadcast Address	172.18.110.191
Usable IP Range	172.18.110.128 – 172.18.110.190

JERUNG – SHARK	
Required Ips	2 IPs
Subnet Size (2^n)	$2^2 = 4 > 2+2$

Subnet Mask	/32 – 2 = /30
Network Address	172.18.110.192/30
Broadcast Address	172.18.110.195
Usable IP Range	172.18.110.193 – 172.18.110.194

JERUNG – TIBURON	
Required Ips	2 IPs
Subnet Size (2^n)	$2^2 = 4 > 2+2$
Subnet Mask	/32 – 2 = /30
Network Address	172.18.110.196/30
Broadcast Address	172.18.110.199
Usable IP Range	172.18.110.197 – 172.18.110.198

SHARK – TIBURON	
Required Ips	2 IPs
Subnet Size (2^n)	$2^2 = 4 > 2+2$
Subnet Mask	/32 – 2 = /30
Network Address	172.18.110.200/30
Broadcast Address	172.18.110.203
Usable IP Range	172.18.110.201 – 172.18.110.202

2. Using the IP addresses calculated, configure the devices with the appropriate information.

Task 2 – Routing Table

1. Paste the current routing table of each router here. There are 2 ways to this – via CLI and via Packet Tracer (PT) tool. Use both ways to show the routing table of all the routers.
   a. CLI
      i. Click on a router. In the prompt, type show ip route. The routing table of the router will be shown, as shown in Figure A.

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
```

Figure A

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.64/26 is directly connected, GigabitEthernet0/0
L       172.18.110.65/32 is directly connected, GigabitEthernet0/0
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.194/32 is directly connected, Serial0/0/0
C       172.18.110.200/30 is directly connected, Serial0/0/1
L       172.18.110.201/32 is directly connected, Serial0/0/1
```

SHARK

```
JERUNG#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 8 subnets, 3 masks
C       172.18.110.0/27 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
C       172.18.110.32/27 is directly connected, GigabitEthernet0/1
L       172.18.110.33/32 is directly connected, GigabitEthernet0/1
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.193/32 is directly connected, Serial0/0/0
C       172.18.110.196/30 is directly connected, Serial0/0/1
L       172.18.110.197/32 is directly connected, Serial0/0/1
```

JERUNG

```
TIBURON#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.128/26 is directly connected, GigabitEthernet0/0
L       172.18.110.129/32 is directly connected, GigabitEthernet0/0
C       172.18.110.196/30 is directly connected, Serial0/1/1
L       172.18.110.198/32 is directly connected, Serial0/1/1
C       172.18.110.200/30 is directly connected, Serial0/1/0
L       172.18.110.202/32 is directly connected, Serial0/1/0
```

TIBURON

b. PT tool

i. Click on the magnifying glass (shown in Figure B), then click on a router, then choose Routing Table. A sample of the result is shown in Figure C.

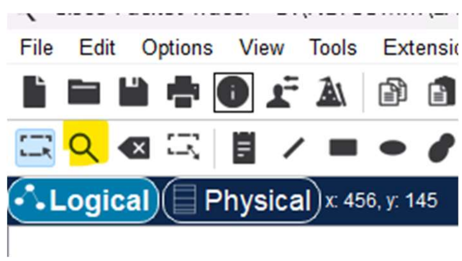
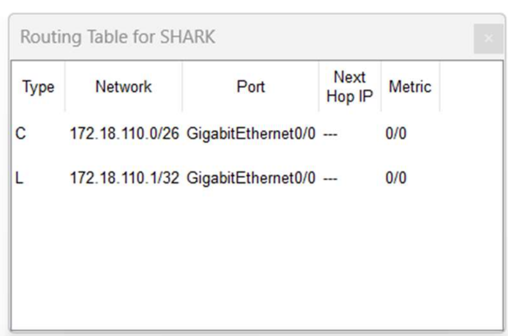


Figure B



Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0

Figure C

Routing Table for JERUNG

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0
L	172.18.110.33/32	GigabitEthernet0/1	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

2. Try to ping PC2 from PC1, paste the results here.

```
C:\>ping 172.18.110.33

Pinging 172.18.110.33 with 32 bytes of data:

Reply from 172.18.110.33: bytes=32 time<1ms TTL=255
Reply from 172.18.110.33: bytes=32 time<1ms TTL=255
Reply from 172.18.110.33: bytes=32 time=1ms TTL=255
Reply from 172.18.110.33: bytes=32 time=1ms TTL=255

Ping statistics for 172.18.110.33:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

3. Try to ping PC4 from PC1, paste the results here.

```
C:\>ping 172.18.110.129

Pinging 172.18.110.129 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Request timed out.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 172.18.110.129:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

4. Explain the reason(s) behind the results.
   The router may be unable to forward the packets to PC4 because the routing information are not configured.
5. What needs to be done to ensure all PCs can ping each other successfully?
   We need to configure the IP address of all PC in the correct subnet.

Task 2 – Routing Configuration

1. Let's start by opening the routing table (using the PT tool) for TIBURON and JERUNG. This is done to show changes to the routing table as configurations are made.

Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

Routing Table for JERUNG

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0
L	172.18.110.33/32	GigabitEthernet0/1	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0

Figure D

2. In router JERUNG, configure the RIP routing protocol as shown in Figure E.

```
JERUNG#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
JERUNG(config)#router rip
JERUNG(config-router)#version 2
JERUNG(config-router)#network 172.18.110.128
JERUNG(config-router)#network 172.18.110.160
JERUNG(config-router)#network 172.18.110.192
JERUNG(config-router)#network 172.18.110.196
JERUNG(config-router)#no auto-summary
JERUNG(config-router)#
```

Figure E

   a. What can you say about the addresses used in the 'network' instructions in Figure E?
   The address in the network instructions is the network address that are connected to router. The devices in different network can communicate along the routes of the router that is in the network instructions.

3. Then configure RIP in TIBURON. All are similar except use the network address. Use the instructions as shown below.

```
TIBURON (config-router) #network 172.18.110.64
TIBURON (config-router) #network 172.18.110.196
TIBURON (config-router) #network 172.18.110.200
```

4. As you may have seen, there are changes in the routing tables of both TIBURON and JERUNG. Paste a copy of these routing tables here.

Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

Routing Table for JERUNG

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0
L	172.18.110.33/32	GigabitEthernet0/1	---	0/0
R	172.18.110.128/26	Serial0/0/1	172.18.110.198	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0
R	172.18.110.200/30	Serial0/0/1	172.18.110.198	120/1

a. What are the changes seen in TIBURON?
There is a new type R in the routing table that have next hop IP of 172.18.110.197 and Metric of 120/1.

b. What are the Networks with type R in TIBURON and JERUNG?
Network with type R in TIBURON:
- 172.18.110.0/27
- 172.18.110.32/27
- 172.18.110.192/30

Network with type R in JERUNG:
- 172.18.110.200/30
- 172.18.110.128/26

c. Ping PC3 from PC1. Was it successful?
Yes.

```
C:\>ping 172.18.110.129

Pinging 172.18.110.129 with 32 bytes of data:

Reply from 172.18.110.129: bytes=32 time=24ms TTL=254
Reply from 172.18.110.129: bytes=32 time=1ms TTL=254
Reply from 172.18.110.129: bytes=32 time=11ms TTL=254
Reply from 172.18.110.129: bytes=32 time=1ms TTL=254

Ping statistics for 172.18.110.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 24ms, Average = 9ms
```

d. Ping PC4 from PC1. Was it successful?
   No.

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Request timed out.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

e. Explain the reasons for your answer.
   PC1 can ping PC3 successfully because RIP is configured in both JERUNG and TIBURON, allowing packets delivery between routes.
   Meanwhile, PC1 Cannot ping PC4 because RIP was not configured in SHARK hence packet cannot be forwarded to PC4.

f. Continue with configuration of RIP in SHARK. Paste your configurations here.

```
SHARK>enable
SHARK#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SHARK(config)#router rip
SHARK(config-router)#version 2
SHARK(config-router)#network 172.18.110.0
```

```
SHARK(config-router)#network 172.18.110.192
SHARK(config-router)#network 172.18.110.200
SHARK(config-router)#no auto-summary
SHARK(config-router)#^Z
SHARK#
%SYS-5-CONFIG_I: Configured from console by console
```

g. Open router SHARK's routing table and paste here.

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.32/27	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
R	172.18.110.128/26	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

h. Try to ping from PC4 to all other PCs in the topology. *Note: Try to ping at least twice to get best results.

PC	Ping result

1	C:\>ping 172.18.110.1 Pinging 172.18.110.1 with 32 bytes of data: Reply from 172.18.110.1: bytes=32 time=1ms TTL=254 Reply from 172.18.110.1: bytes=32 time=1ms TTL=254 Reply from 172.18.110.1: bytes=32 time=1ms TTL=254 Reply from 172.18.110.1: bytes=32 time=1ms TTL=254 Ping statistics for 172.18.110.1: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 1ms, Average = 1ms C:\>ping 172.18.110.1 Pinging 172.18.110.1 with 32 bytes of data: Reply from 172.18.110.1: bytes=32 time=22ms TTL=254 Reply from 172.18.110.1: bytes=32 time=1ms TTL=254 Reply from 172.18.110.1: bytes=32 time=1ms TTL=254 Reply from 172.18.110.1: bytes=32 time=1ms TTL=254 Ping statistics for 172.18.110.1: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 22ms, Average = 6ms C:\>
2	C:\>ping 172.18.110.33 Pinging 172.18.110.33 with 32 bytes of data: Reply from 172.18.110.33: bytes=32 time=27ms TTL=254 Reply from 172.18.110.33: bytes=32 time=2ms TTL=254 Reply from 172.18.110.33: bytes=32 time=18ms TTL=254 Reply from 172.18.110.33: bytes=32 time=7ms TTL=254 Ping statistics for 172.18.110.33: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 2ms, Maximum = 27ms, Average = 13ms C:\>ping 172.18.110.33 Pinging 172.18.110.33 with 32 bytes of data: Reply from 172.18.110.33: bytes=32 time=24ms TTL=254 Reply from 172.18.110.33: bytes=32 time=14ms TTL=254 Reply from 172.18.110.33: bytes=32 time=1ms TTL=254 Reply from 172.18.110.33: bytes=32 time=1ms TTL=254 Ping statistics for 172.18.110.33: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 24ms, Average = 10ms

<table>
<tr><td>3</td><td>C:\>ping 172.18.110.129

Pinging 172.18.110.129 with 32 bytes of data:

Reply from 172.18.110.129: bytes=32 time=22ms TTL=254
Reply from 172.18.110.129: bytes=32 time=24ms TTL=254
Reply from 172.18.110.129: bytes=32 time=1ms TTL=254
Reply from 172.18.110.129: bytes=32 time=1ms TTL=254

Ping statistics for 172.18.110.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 24ms, Average = 12ms

C:\>ping 172.18.110.129

Pinging 172.18.110.129 with 32 bytes of data:

Reply from 172.18.110.129: bytes=32 time=23ms TTL=254
Reply from 172.18.110.129: bytes=32 time=1ms TTL=254
Reply from 172.18.110.129: bytes=32 time=1ms TTL=254
Reply from 172.18.110.129: bytes=32 time=14ms TTL=254

Ping statistics for 172.18.110.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 23ms, Average = 9ms</td></tr>
</table>

Task 3 – Routing Update

1. Let's try a little experiment. Change the IP addresses of router TIBURON interface G0/0 to 192.168.1.1/24.
    a. This means that the subnet has changed. Find the new Network address of this subnet. Network Address is: 192.168.1.0/24
    b. As this change happens, PC3 must also have different IP address, subnet mask and gateway address. What will it be?

PC3	Info
IP address	192.168.1.254
Subnet Mask	255.255.255.0
Gateway Address	192.168.1.1

    c. After this change, can PC4 and PC1 ping PC3?
    PC4 cannot ping PC3

```
C:\>ping 192.168.1.254

Pinging 192.168.1.254 with 32 bytes of data:

Reply from 172.18.110.65: Destination host unreachable.
Reply from 172.18.110.65: Destination host unreachable.
Reply from 172.18.110.65: Destination host unreachable.
Reply from 172.18.110.65: Destination host unreachable.

Ping statistics for 192.168.1.254:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
```

PC1 cannot ping PC3

```
C:\>ping 192.168.1.254

Pinging 192.168.1.254 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 192.168.1.254:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

d. Copy and paste the routing tables for both SHARK and TIBURON here.

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.32/27	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.64/26	Serial0/1/0	172.18.110.201	120/1
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.192/30	Serial0/1/0	172.18.110.201	120/1
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0
C	192.168.1.0/24	GigabitEthernet0/0	---	0/0
L	192.168.1.1/32	GigabitEthernet0/0	---	0/0

e. Referring to the routing table, explain your findings.
TIBURON is directly connected to the 192.168.1.0/24 subnet, but SHARK is not connected to it.

f. What is your next move to ensure end-to-end connectivity (i.e. all PCs can ping each other successfully)?
Configure RIP from TIBURON.

g. Show your configurations in TIBURON to ensure end-to-end connectivity.

```
TIBURON(config)#router rip
TIBURON(config-router)#version 2
TIBURON(config-router)#network 192.168.1.0
TIBURON(config-router)#network 172.18.110.196
TIBURON(config-router)#network 172.18.110.200
TIBURON(config-router)#no auto-summary
TIBURON(config-router)#^Z
TIBURON#
%SYS-5-CONFIG_I: Configured from console by console
```

h. To ensure end-to-end connectivity, ping to all the PCs from PC3.

PC	Ping result
1	C:\>ping 172.18.110.1 Pinging 172.18.110.1 with 32 bytes of data: Reply from 172.18.110.1: bytes=32 time=31ms TTL=254 Reply from 172.18.110.1: bytes=32 time=20ms TTL=254 Reply from 172.18.110.1: bytes=32 time=28ms TTL=254 Reply from 172.18.110.1: bytes=32 time=1ms TTL=254 Ping statistics for 172.18.110.1: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 31ms, Average = 20ms
2	C:\>ping 172.18.110.33 Pinging 172.18.110.33 with 32 bytes of data: Reply from 172.18.110.33: bytes=32 time=1ms TTL=254 Reply from 172.18.110.33: bytes=32 time=12ms TTL=254 Reply from 172.18.110.33: bytes=32 time=11ms TTL=254 Reply from 172.18.110.33: bytes=32 time=22ms TTL=254 Ping statistics for 172.18.110.33: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 22ms, Average = 11ms

<table>
<tr>
<td>4</td>
<td>C:\>ping 172.18.110.65

Pinging 172.18.110.65 with 32 bytes of data:

Reply from 172.18.110.65: bytes=32 time=1ms TTL=254
Reply from 172.18.110.65: bytes=32 time=16ms TTL=254
Reply from 172.18.110.65: bytes=32 time=1ms TTL=254
Reply from 172.18.110.65: bytes=32 time=12ms TTL=254

Ping statistics for 172.18.110.65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 16ms, Average = 7ms</td>
</tr>
</table>

REFLECTION

What have you learned in this task?

I have learned about planning and assigning IP addresses to routers. It makes sure the effective communication between devices by dividing networks into subnets, assign IP addresses without overlap and implement routing with RIP. I also learned some ways to debug the issues with my connections by pinging to identify the problem. Lastly, I learned about noticing the change in routing tables.